AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

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AWS Academy Certified Educator

5.Configure Apache Web Server on Linux Instance in AWS

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Objective

EC2 (Elastic Compute Cloud) is a core serviceprovided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Linux instance is a virtual machine running the Linux operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, Apache,FileServer etc.) in the cloud without having to manage physical hardware.

Use Cases for EC2 Linux Instances:

- Web and Application Hosting: Run Apache web servers or other Linux-based web applications in the cloud.
- Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.
- Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.
- Development and Testing: Create development environments with linux OS to test php,MySql or other Linux-based applications.

Used Services in AWS

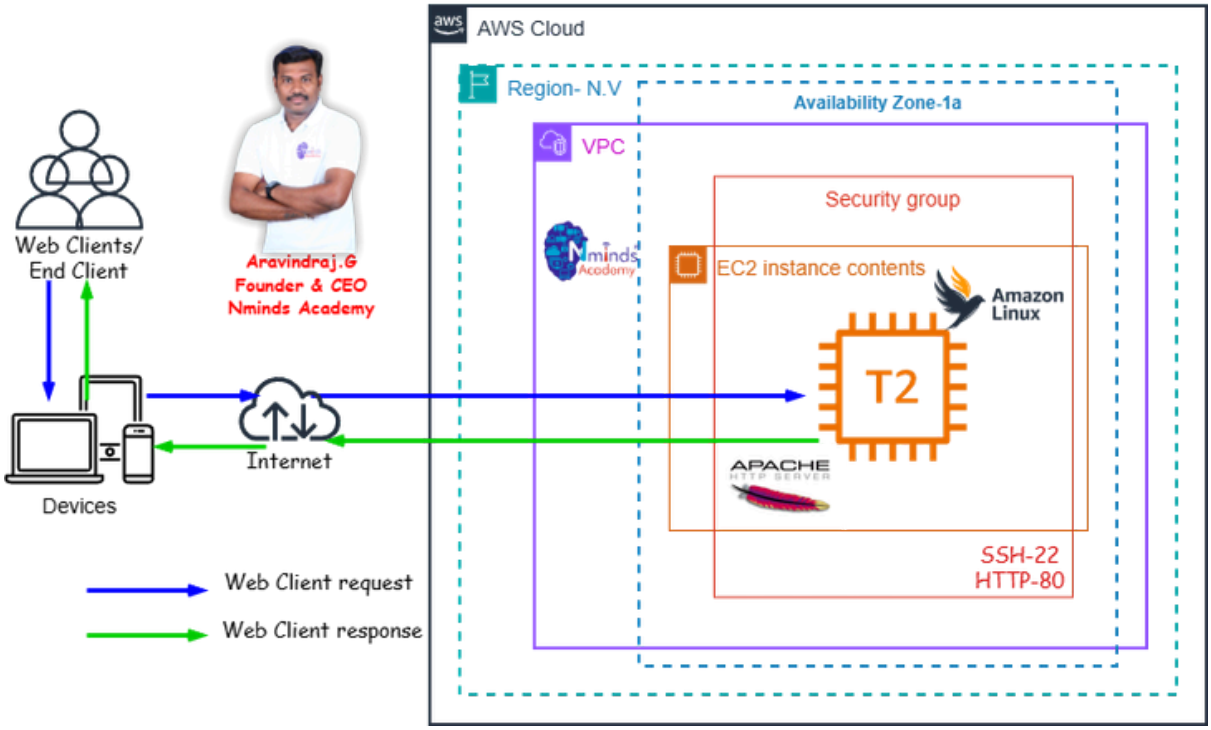
Amazon EC2

Used Services in Linux Server

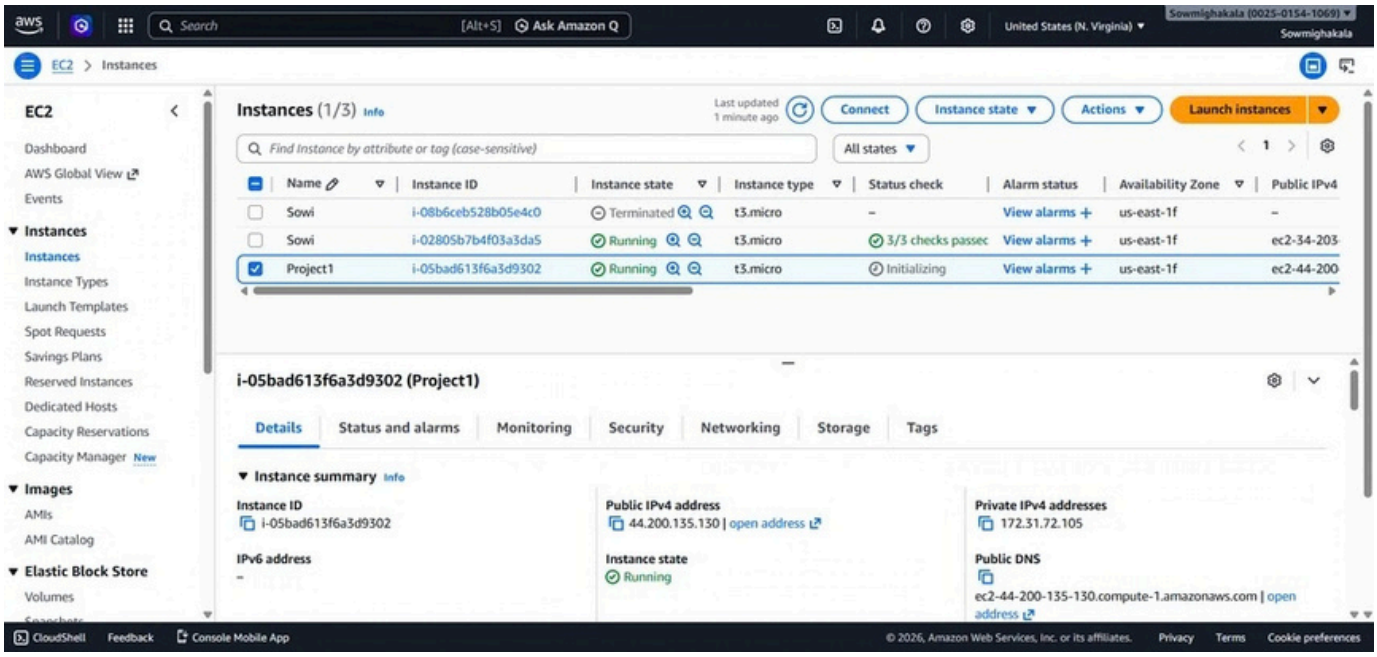
Apache Webserver

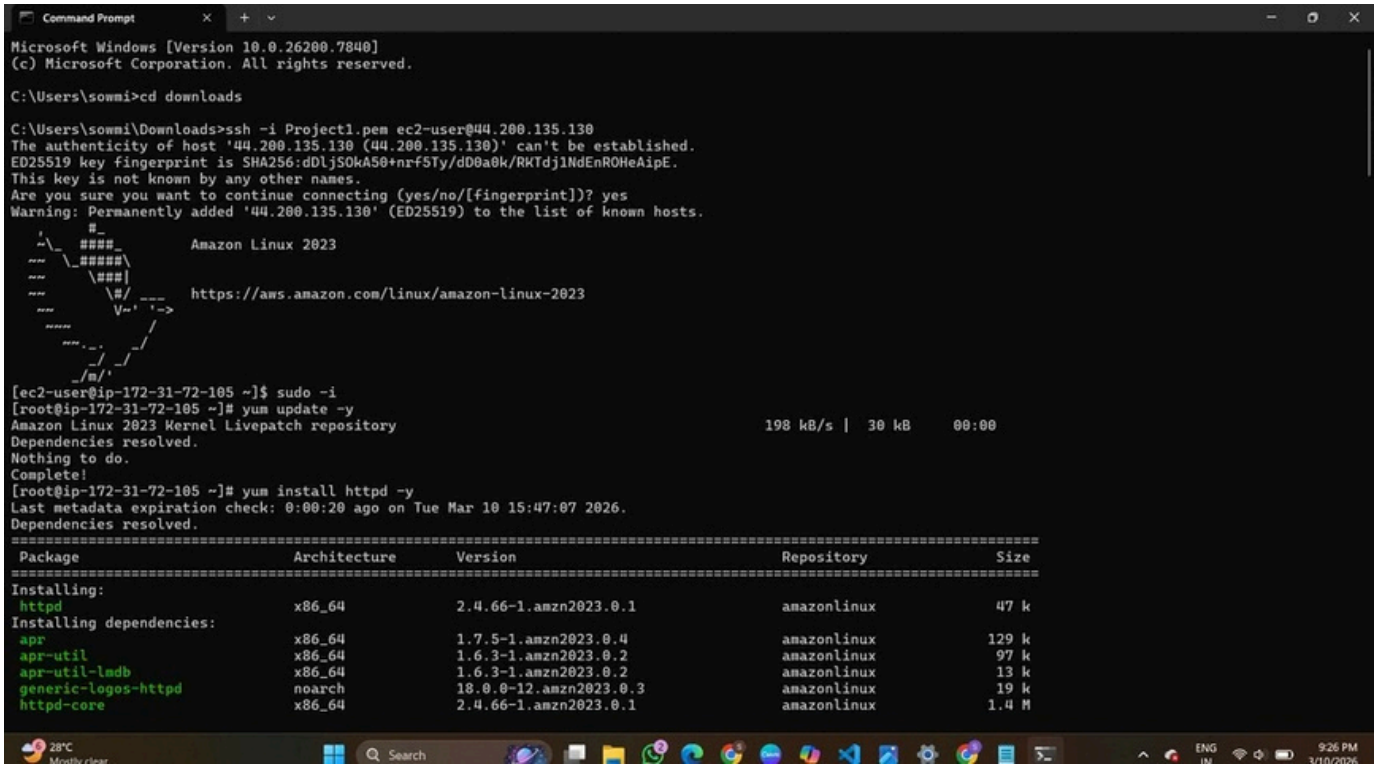
Topology

How to Configure the Apache web server in Linux Server with AWS EC2



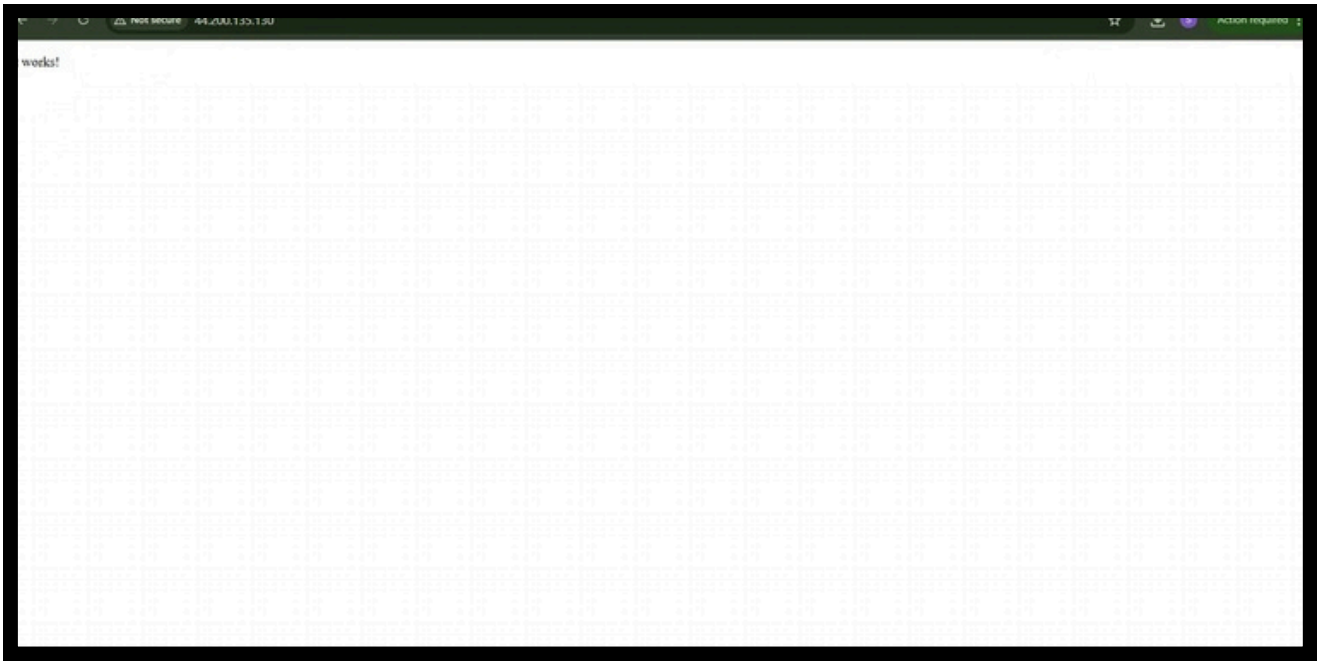
Execution Tasks:





Execution Tasks:

```
Command Prompt
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch          httpd-2.4.66-1.amzn2023.0.1.x86_64
  httpd-core-2.4.66-1.amzn2023.0.1.x86_64                    httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
  httpd-tools-2.4.66-1.amzn2023.0.1.x86_64                   libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  mailcap-2.1.49-3.amzn2023.0.3.noarch                       mod_http2-2.0.27-1.amzn2023.0.3.x86_64
  mod_lua-2.4.66-1.amzn2023.0.1.x86_64

Complete!
[root@ip-172-31-72-105 ~]# systemctl status httpd
○ httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: inactive (dead)
       Docs: man:httpd.service(8)
[root@ip-172-31-72-105 ~]# systemctl start httpd
[root@ip-172-31-72-105 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: active (running) since Tue 2026-03-10 15:48:09 UTC; 5s ago
       Docs: man:httpd.service(8)
   Main PID: 25423 (httpd)
     Status: "Started, listening on: port 80"
      Tasks: 177 (limit: 1067)
     Memory: 13.4M
        CPU: 87ms
     CGroup: /system.slice/httpd.service
             ├─25423 /usr/sbin/httpd -DFOREGROUND
             ├─25448 /usr/sbin/httpd -DFOREGROUND
             ├─25449 /usr/sbin/httpd -DFOREGROUND
             ├─25461 /usr/sbin/httpd -DFOREGROUND
             └─25478 /usr/sbin/httpd -DFOREGROUND

Mar 10 15:48:09 ip-172-31-72-105.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Mar 10 15:48:09 ip-172-31-72-105.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Mar 10 15:48:09 ip-172-31-72-105.ec2.internal httpd[25423]: Server configured, listening on: port 80
[root@ip-172-31-72-105 ~]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@ip-172-31-72-105 ~]# cd /var/www/html
[root@ip-172-31-72-105 html]# ls
[root@ip-172-31-72-105 html]# vi index.html
"index.html" [New] 113L, 1954B written
[root@ip-172-31-72-105 html]# client_loop: send disconnect: Connection reset
```

```
root@ip-172-31-72-105:~
# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

PubkeyAuthentication no

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile      .ssh/authorized_keys

#AuthorizedPrincipalsFile none


# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# Explicitly disable PasswordAuthentication. By presetting it, we
# avoid the cloud-init set_passwords module modifying sshd_config and
# restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no

# Change to no to disable s/key passwords
#KbdInteractiveAuthentication yes

# Kerberos options
#KerberosAuthentication no
-- INSERT --                                                    65,27         34%
```

Execution Tasks: